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Research Paper



Thermal performance analysis of supporting column ultra-thin vapor chamber with graded microgroove

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ABSTRACT

To analyze the thermal behavior of a graded microgroove ultra-thin vapor chamber with supporting column, a three-dimensional steady-state numerical model coupled with graded capillary force network, which involves hydraulic and heat transfer on hydraulic and heat transfer performance on ultra-thin vapor chamber with different supporting column sizes. This model considers the influence of supporting column with varying diameters and intervals on liquid flow and temperature distribution, as well as the effects of a graded capillary force network on heat transfer limit of ultra-thin vapor chamber. The numerical results show that the additional heat transfer provided by the central support column causes an outward shift in the saturation temperature of vapor, creating a new liquid reflux path at the condensation surface. The accuracy of the model is verified by the preparation of ultra-thin vapor chamber with a graded microgroove a graded microgroove. Compared with the experimental results on conventional microgrooves, graded microgrooves can improve the heat transfer limit by 29.3%. According to the theoretical analysis model, the graded capillary force network can effectively resist incremental increases in vapor pressure drop, enhancing the efficiency of gas–liquid circulation of ultra-thin vapor chamber.

1. Introduction

Ultra-thin vapor chambers (UTVCs) are high thermal conductivity devices that leverage the vapor–liquid phase change principle, making them essential for thermal management in portable electronics such as smartphones, tablets, AR/VR devices [1], smartwatches, and notebooks. With the increasing power density and the demand for miniaturization in these devices, UTVCs play a critical role in dissipating heat effectively [2,3]. To enhance their thermal performance, it's important to optimize the design of UTVCs, including the wick structure, support columns, and working fluids [4]. Studies have shown that UTVCs can achieve higher heat transfer efficiency and reduced size and weight [5]. For instance, Huang et al. [6] developed a 1.26 mm thick UTVC with a thermal resistance as low as 0.197 ℃/W, capable of handling up to 50 W of heat. Long et al. [7] utilized nanosecond laser surface texturing to create wick structures, resulting in a 2 mm thick vapor chamber with a thermal resistance of 0.137 ℃/W under 120 W of heat. Xie et al. [8] employed ultrafast laser technology to fabricate a gas–liquid partitioned microgroove structure, achieving a minimum thermal resistance of 0.082 ℃/W and a maximum heat transfer coefficient of 30,652 W/(m^2·K) in their UTVC. It has been shown that the heat transfer limit of UTVC can be enhanced by gradient capillary force.

The gradient micro-nano structure inspired by leaf vein [9,10], cactus [11], and peristome surface of Nepenthes [12] could enhance the capillary force of the wick by utilizing the meniscus formed in narrow corners of liquid passing through grooves. As the UTVC thickness decreases, it is necessary to introduce support columns to avoid the significant decrease in heat transfer performance of UTVC caused by wall collapse deformation caused by atmospheric pressure [13]. However, among the available literature, there is a scarcity of theoretical analyses exploring the effects of temperature, pressure, and velocity distributions caused by pure copper support columns coupled with the graded microgroove on the thermal performance of UTVCs, specifically the intricate coupling between velocity, temperature, and pressure at the liquid–vapor interface. This challenge arises due to the lack of an effective quantitative analysis technique for examining the internal flow processes of UTVCs [14]. Therefore, several numerical models have been proposed to systematically investigate the thermodynamic and flow characteristics of UTVCs.

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Nomenclature

A	area (m^2)
A_{hs}	area of heat source (m^2)
Cp	specific heat (J/kg·K)
d	width (m)
D	diameter (m)
D_h	hydraulic diameter (m)
(f_{Re})	friction coefficient (−)
h	heat transfer coefficient (W/(m^2·K))
h_{fg}	latent heat of vaporization (J/kg)
I_{sp}	interval of support columns (m)
k	thermal conductivity W/(m·K)
k_{eff}	effective thermal conductivity W/(m·K)
K	permeability (−)
L	Length (m)
m“	interface mass flux (kg/s)
n	normal vector
N	number of units at the condenser (−)
p	pressure (Pa)
P	power (W)
q	heat transfer rate (W/m^2)
r	radius (m)
r_{ite}	iterative residual (−)
R	thermal resistance (K/W)
R_g	universal gas constant (J/mol·K)
S_t	momentum source term (N·m^3)
t	time (s)
T	temperature (K)
$T_{1\text{-}8}$	temperature of thermocouple (K)
T_{ave}	average temperature (K)
T_{cool}	temperature of cooling water (K)
u	velocity in groove (m/s)
U_T	temperature uniformity (K)
V	volume (m^3)
V“	velocity of liquid (m/s)
w	interval between adjacent groove (m)

Greek Symbols

α	contact angle ()
α_{ac}	accommodation coefficient (−)
δ	thickness (m)
ε	porosity (%)
η	filling rate (%)
μ	dynamic viscosity (Pa·s)
ρ	density of liquid (kg/m^3)
σ	surface tension (N/m)
τ	shearing force (Pa)
ϕ	numerical result (−)
$\Delta_{li,j}$	normal length (m)
Δ_p	pressure difference (N/m^3)
Δ_T	temperature difference (K)

Subscripts

c	capillary
co	condensation
cw	condenser wick
cs	copper shell except for wick
e	evaporation
ew	evaporation wick
i	unit cell element
im	intrinsic menisci
l	liquid
max	maximum
min	minimum
sat	saturation
sl	solid–liquid interface
sp	support support columns
sv	solid–vapor interface
v	vapor
vc	vapor chamber
vl	vapor–liquid interface
w	solid wall

The thermodynamic and flow characteristics within a UTVC exhibit a dual nature of independence and interrelation [15]. The numerical analyses of thermodynamics cannot overlook the influence of support columns on the thermal performance of UTVCs [16]. Factors such as the effect of vapor flow on supporting columns shape [17], the number of supporting columns [18], the relationship between channel sizes and shell support [19] are critical for accurate analysis. Additionally, numerical methods for modeling flow characteristics within wicks typically rely on Darcy's law for capillary forces in reticulated copper powders. For conventional groove wicks, the Young-Laplace equation is mainly used to calculate the hydraulic diameter, the shear force at the interface and the capillary force in the axial groove. In contrast to a one-dimensional steady-state equivalent model, the complexity of a pressure–temperature coupled 3D transient mathematical model requires a simplification approach for the thermal and hydraulic numerical solutions in the inner region of UTVC, spanning various levels and classifications [20,21]. Xin et al. [22] proposed a novel graded-groove wick design and established an accurate mathematical model for mini-grooves flat heat pipes (heat transfer in the horizontal direction). The effective thermal conductivity of the proposed heat pipe is increased by 12.4 %. However, a three-dimensional gradient capillary force homogeneous heat plate model for heat transfer in the vertical direction has not been established. Thus, this study establishes a three-dimensional steady-state model coupling support columns with graded micro-groove developed to analyze and optimize the heat transfer performance of UTVCs, which are based on the fundamental principles governed by the momentum equation, the continuity equation, and the energy equation to solve the Navier-Stokes equation [23–25].

Numerical simulations are conducted to systematically investigate the hydraulic and heat transfer performance of UTVCs with varying parameters of support columns and types of microgrooves. These simulations aim to reveal the additional thermal conductivity pathway provided by the copper columns and the enhancement effect of graded microgrooves on UTVC performance.

2. Numerical Descriptions

2.1. Assumptions

As shown in Fig. 1(a), the computation of UTVC with copper columns is divided into four parts: the walls, the micropillars, the wicks, and the vapor core. The walls and micropillars are modeled as solid copper, the vapor core as single-phase flow (vapor), and the wicks as single-phase flow (liquid). To facilitate simulation analysis, one quarter of the UTVC is used for numerical analysis, as the actual UTVC is symmetric about the heat source region (denoted by the yellow dashed box in Fig. 1 (b)). The assumptions are listed as follows [26,27]:

(1) Compressible laminar flow.

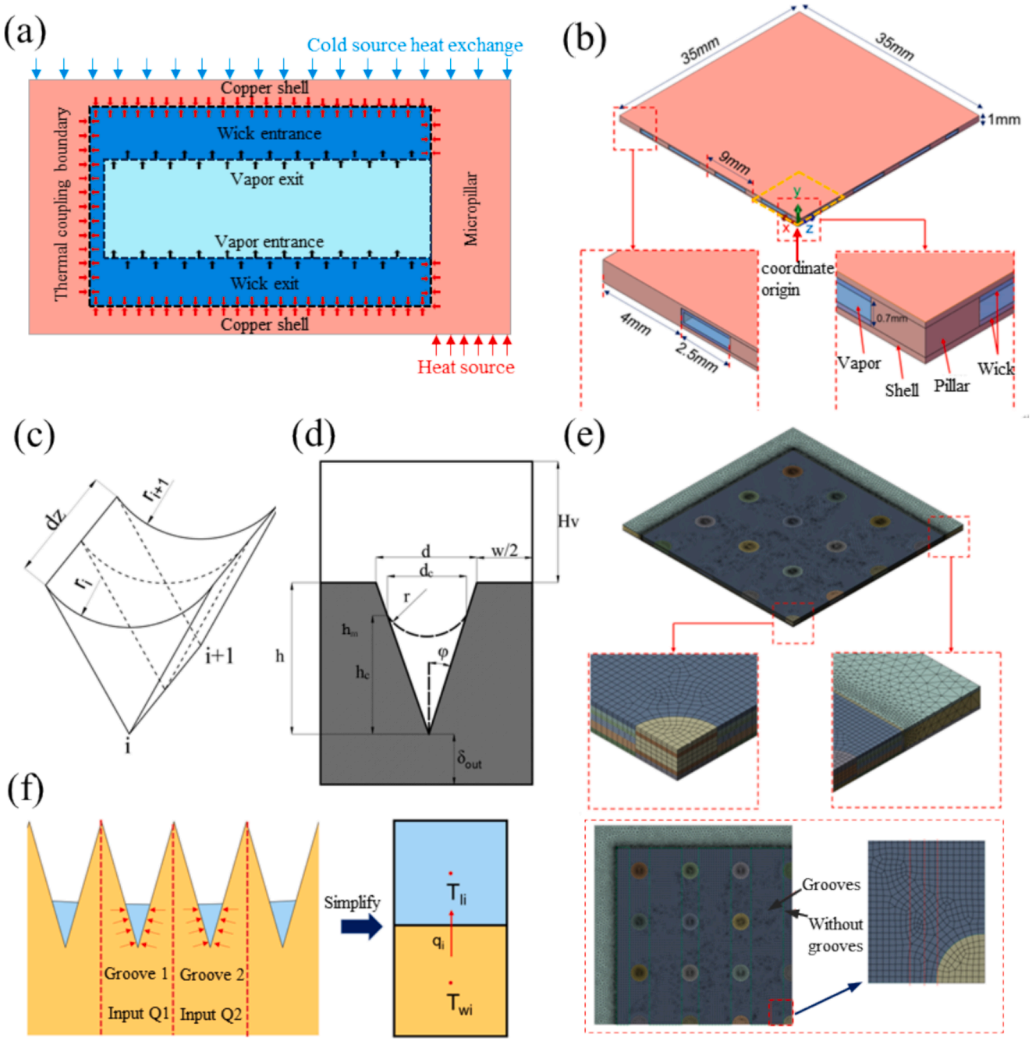


Fig. 1. (a) Schematic of boundary conditions for the UTVC model. (b) Diagram of geometric model with UTVC. (c) Illustration of the calculation unit for liquid in a microgroove. (d) Schematic of dimensions in microgroove. (e) Meshing principle at the graded microgroove coupled with support columns UTVC model. (f) Simplification of capillary-thermal model in microgroove.

(2) Thin film evaporation occurs exclusively at the liquid–vapor interface, and dry-out of the liquid does not occur.
(3) The edge of the wall area is considered adiabatic, and no heat leakage.
(4) The liquid completely infiltrates the wick, with the capillary radius varying solely along the microgroove direction.
(5) The vapor core layer is assumed to be isothermal and saturated with vapor.
(6) Heat exchange within the vapor core layer and the effects of gravity are neglected.

2.2. Governing equations

The fluid pressure and velocity distributions within the wicks and vapor core layer are determined by incorporating the stress tensor from constitutive equations into the momentum equation, which is then coupled with Navier-Stokes equations adapted for compressible fluids [28]. Simultaneously, temperature distributions in the wall layer and support columns are calculated using the principles of Fourier's law.

Based on the above assumptions, the continuity equation, momentum equation, and ideal gas equation for the vapor core region are represented as Eq (1)–(3), respectively:

$$\frac{\partial \rho}{\partial t} + \nabla \bullet (\rho \boldsymbol{V_v}) = 0 \tag{1}$$

$$\rho_v \boldsymbol{V_v} \nabla \boldsymbol{V_v} = -\nabla p_v + \mu_v \nabla^2 \boldsymbol{V_v} \tag{2}$$

$$p = \rho R T_v \tag{3}$$

The continuity equation, energy equation, and the capillary force mass source term for the wicks are represented as Eq (4), Eq (5), and Eq (6) respectively:

$$\frac{\partial \rho}{\partial t} + \nabla \bullet (\rho \boldsymbol{V_l}) = 0 \tag{4}$$

$$\rho_l \boldsymbol{V_l} \nabla \boldsymbol{V_l} = -\nabla p_l + \mu_l \nabla^2 \boldsymbol{V_l} \tag{5}$$

$$\frac{\partial (\rho T_l)}{\partial t} + \nabla \bullet (\rho \boldsymbol{V_l} T_l) = \nabla \bullet \left(\frac{k}{c_p} \nabla T_l \right) + S_T \tag{6}$$

In the wall, the governing equation is:

$$\rho C_p \frac{\partial T_w}{\partial t} = -\nabla \bullet (k \nabla T_w) + \boldsymbol{q} \tag{7}$$

2.3. Initial and boundary conditions

2.3.1. Wall boundary

There are three types of conditions for heat transfer applied on the exterior walls. The heating area (heat source region) is applied by a constant input heat flux (q_i), as shown in (8):

$$q_i = P/A_{source} \tag{8}$$

The condenser wall is subjected to convective heat transfer with a constant and uniform heat transfer coefficient (h_c), as shown in Eq (9), and Eq (10) [29]:

$$q_i = (T_i - T_{cool})^* h_c \tag{9}$$

$$T_{hs} = T_{cool} + q(\frac{\delta_t}{k} + \frac{1}{h_c}) \tag{10}$$

Where P is the heating power, A_s is the heating area, δ_t is the thickness of copper plate, T_{hs} is the evaporation center temperature, T_{cool} is the condensation temperature, (set at 40 °C), h_{co} is the heat transfer coefficient, with a value of 5102 W/(m^2·℃) in this study.

The other walls are assumed to be adiabatic [30]:

$$\boldsymbol{V_i} = 0 \tag{11}$$

$$\boldsymbol{q_i} = 0 \tag{12}$$

The effective thermal conductivity of support columns (k_{wc}) is expressed by Eq. (13).

$$k_{wc}\frac{\partial T_{wc}}{\partial n} = k_{eff}\frac{\partial T_l}{\partial n} \tag{13}$$

2.3.2. Liquid-vapor interfaces

The conservation of mass at the liquid–vapor interface, the mass flow rate is applied:

$$\dot{m}_i = \frac{k_i A_i (T_v - T_i)}{h_{fg}\Delta l_i/2} \tag{14}$$

Where k_i is the unit thermal conductivity, A_i is the unit boundary area, T_v is the vapor temperature, h_{fg} is the latent heat of working fluid, Δl_i is the unit normal length, T_i is the unit temperature.

The vapor temperature in vapor core layer is governed by Eq. (15):

$$T_v = \frac{\sum_{i=0}^{ew,M}\frac{k_i A_i}{\Delta l}T_i + \sum_{j=0}^{cw,N}\frac{k_i A_j}{\Delta l}T_i}{\sum_{i=0}^{ew,M}\frac{k_i A_i}{\Delta l} + \sum_{j=0}^{cw,N}\frac{k_i A_j}{\Delta l}} \tag{15}$$

where the ew and the cw are abbreviations for the evaporation wick and the condensation wick, respectively; M and N are the number of boundary cells for the evaporation wick and the condensation wick, respectively.

2.4. Governing equations of capillary force for groove

The capillary radius of microgrooves and the distribution of capillary force are determined using the Young-Laplace equation (the microgrooves are aligned along the z axis). The calculation of capillary force in the microgrooves is achieved by solving the fourth-order Runge-Kutta method, and the domain is discretized for finite volume computations in FLUENT. The computational unit for discretization of liquid in microgrooves is illustrated in Fig. 1(c). The dimensions of the microgroove are shown in Fig. 1(d), where δ_c is the height of the meniscus, δ is groove depth, δ_{dc} the width of the meniscus, δ_{out} the distance of microgroove from the wall, H_v is the height of the vapor chamber, the contact angle α is 10°.

The microgroove liquid velocity at liquid–vapor interface is set as follows:

$$\rho\frac{d(AV)}{dz} = \frac{1}{h_{fg}}\frac{dq(z)}{dz} \tag{16}$$

As shown in Fig. 1(f), the heat conduction of the liquid inside microgroove is simplified, and the unit heat flux q_i and the effective thermal conductivity of the microgroove wicks k_{eff} can be expressed by (17) to (18).

$$q_i = \frac{k_{eff}A_i(T_{ei} - T_{li})}{\Delta l/2} \tag{17}$$

$$k_{eff} = \frac{k_l[(k_l + k_w) - (1-\varepsilon)(k_l - k_w)]}{(k_l + k_w) + (1-\varepsilon)(k_l - k_w)} \tag{18}$$

Where ε represents the porosity of the microgrooves, assumed to be 0.5.

Due to the absence of thickness at the microgroove/vapor interface, the interface pressure within the microgroove can be calculated using principles of energy conservation:

$$\frac{d(\rho_l A_l u_l^2)}{dz} + A_l\frac{dp_l}{dz} = |\tau_{il}|\frac{dA_{il}}{dz} + |\tau_{lw}|\frac{dA_{lw}}{dz} \tag{19}$$

$$\alpha_v\frac{d(\rho_l A_v u_v^2)}{dz} + A_v\frac{dp_v}{dz} = -|\tau_{iv}|\frac{dA_{iv}}{dz} - |\tau_{vw}|\frac{dA_{vw}}{dz} \tag{20}$$

where α_v is the accommodation coefficient, which is mainly determined experimentally and the value adopted in the simulation is 0.03.

The flow of vapor and liquid is influenced by mutual shear, and the intrinsic friction coefficient of the groove is employed in this model to calculate the shear between vapor and liquid flow:

$$\frac{2\rho_l u_l d(u_l)}{dz} + \frac{\rho_l u_l^2 d(A_l)}{A_l dz} + \frac{dp_l}{dz} = \frac{4|\tau_l|}{D_{h,l}} \tag{21}$$

$$\frac{2\alpha_v\rho_v u_v d(u_v)}{dz} + \frac{\alpha_v\rho_v u_v^2 d(A_v)}{A_v dz} + \frac{dp_v}{dz} = \frac{4|\tau_v|}{D_{h,v}} \tag{22}$$

The liquid shear τ_l and the vapor shear τ_v are governed by Eq (23) and Eq (24), respectively.

$$\tau_l = \frac{1}{2}\frac{(fRe)_l\mu_l u_l}{D_{h,l}} \tag{23}$$

$$\tau_v = \frac{1}{2}\frac{(fRe)_v\mu_v u_v}{D_{h,v}} \tag{24}$$

The distribution of capillary radius is obtained from the differential form of the Young-Laplace equation, influenced by the pressure of vapor and liquid.

$$\frac{dp_v}{dz} - \frac{dp_l}{dz} = -\frac{\sigma}{r^2}\frac{dr}{dz} \tag{25}$$

The graded microgroove with capillary force distribution is expressed by the discrete Laplace equation.

$$\Delta p_c = \frac{\sigma}{r_n} - \frac{\sigma}{r_{n+1}} \tag{26}$$

Fig. 1(e) illustrates the grid of capillary forces within the graded

Table 1
Physical properties of materials.

Material	Thermal conductivity (W/m/K)	Density (kg/m^3)	Constant-pressure heat capacity (kg/m/s)	Viscosity (kg/m/s)
copper	387.6	8978	381	–-
liquid	0.6	–-	4182	1.03*10^{-3}
vapor	0.0261	–-	4182	1.34*10^{-3}

microgroove. The physical properties of materials are shown in Table 1. A vapor chamber model is developed with non-uniform step lengths, whereas a microgroove wick model used uniform steps. The discrete temperature distribution of individual grid units is adjusted to create a continuous approximation. Wall and fluid temperatures, uniformly spaced, are subsequently interpolated into the graded microgroove wick model. Importantly, the microgrooves are constrained by support columns, confining the computation of capillary forces to the region outlined by the green dashed line in Fig. 1(e). The numerical analysis procedure is shown in Fig.S1. The Mesh methods and parameter and Convergence condition of the source term are shown in Table S1 and Table S2.

This study focuses on a coupled capillary force model tailored for the graded microgroove configuration. However, the lack of alignment between the evaporation and condensation locations on the soaking plate disrupts the continuity of capillary force within the microgrooves, posing challenges for model convergence. Consequently, this article specifically introduces the capillary force model for evaporation into the three-dimensional steady-state UTVC model for numerical analysis.

3. Results and discussion

3.1. Capillary force of different V-type microgroove

Fig. 2 illustrates the dimensions and capillary force distributions of a conventional V-type microgroove wick (CV-wick) and a gradient V-type microgroove wick (GV-wick). In Fig. 2(a), the CV-wick features dimensions of 70 μm width, 105 μm depth, and 70 μm spacing between adjacent microgrooves. Contrarily, in Fig. 2(b), the GV-wick maintains a depth of 105 μm but exhibits a width that increases from 70 μm at the groove center to 100 μm at the groove edge. This design of the GV-wick enhances capillary transport capacity by providing additional Laplace forces towards the evaporation center. Comparing the capillary force distributions between CV-wick and GV-wick shown in Fig. 2(c) and (d), the maximum capillary force of the CV-wick is 1670 N/m^3, whereas that of the GV-wick is 3720 N/m^3. This enhancement in capillary force of the GV-wick, which is 2.22 times greater than that of the CV-wick, results from increased evaporation induced by higher heat flux. The GV-wick also ensures there is additional force at the evaporation center, facilitating fluid replenishment. In this study, the graded microgroove prepared using ultrafast laser technology are simpler and significantly enhanced the capillary force compared to grooves wick in slope type [20]. To visualize the three-dimensional distribution of capillary force for the CV-wick, Fig. 2(e) presents a grid arrangement of the UTVC coupled capillary force model. The plot reveals a slight decline in capillary force along the z-axis, indicating potential challenges in achieving uniform phase transition away from the heat source regions. The numerical results indicated that the gradient micro-nano structure (graded microgroove) is capable of significantly enhancing the maximum capillary force of the wick near the edge of heat source regions, thereby furnishing a robust theoretical framework for the development of high-performance UTVCs in this study.

3.2. Thermal characteristics of UTVC with support column

In this study, it is crucial to consider the influence of support columns on thermal performance of UTVCs when incorporating a high capillary force GV-wick. In order to clarify the thermal characteristics of UTVCs with support column, the heating condition at an input heat flux of 75,000 W/m^2 is analyzed. Fig. 3 shows the steady-state temperature distribution of copper plate, the void-UTVC, and the diameter 3 mm support columns UTVC(SC-UTVC). Fig. 3(a2-c2) and (a3-c3) illustrate their respective temperature profiles at the evaporation and condensation wicks. Temperature gradients originate from the evaporative region (indicated by the red dashed box representing 1/4 of the heat source). Significant temperature variations are observed at the SC-UTVC between the wicks and support structures. Conversely, temperature gradients on the copper plate are less pronounced towards the evaporation surface, with the highest temperature reaching 48.82 ℃. This temperature is 7.64 ℃ and 4.02 ℃ lower than in the void-UTVC and SC-UTVC, respectively. The abbreviations for the different UTVCs have been detailed as shown in Table 2.

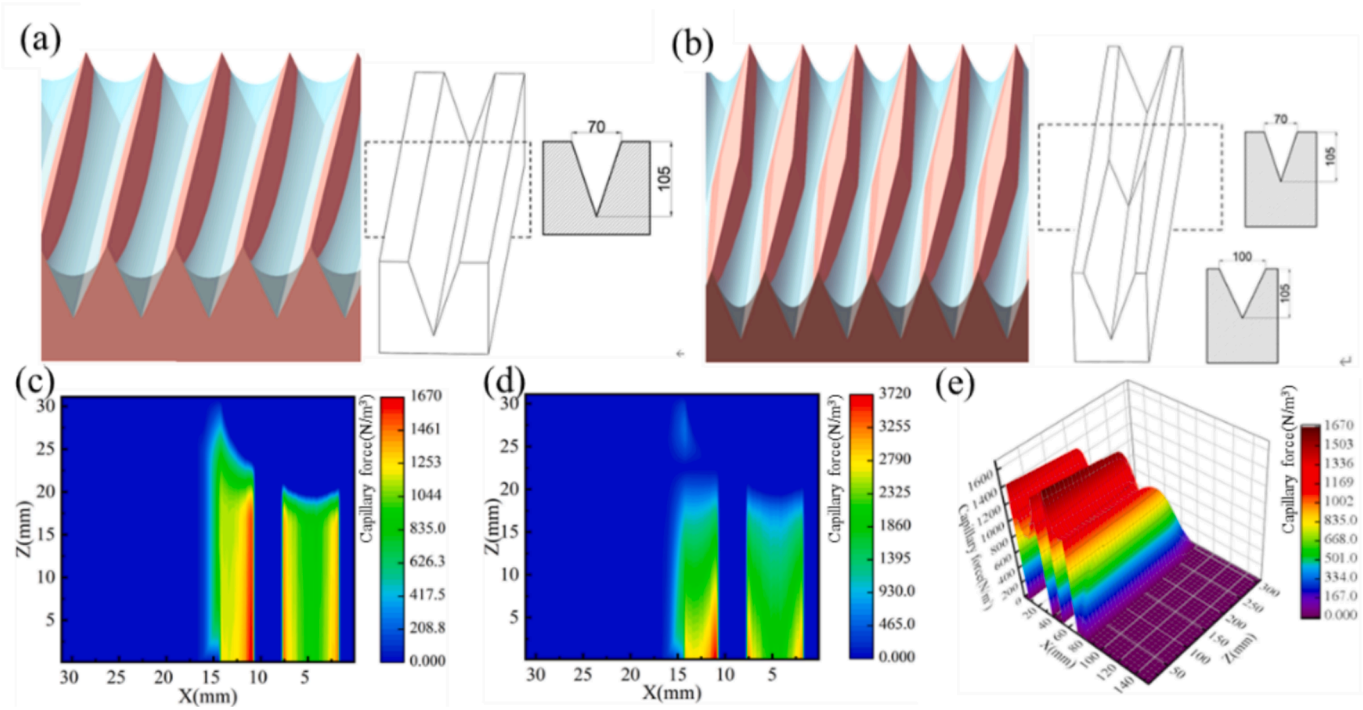


Fig. 2. (a, b) Schematic dimensions of CV-wick and GV-wick, respectively; (c) the maximum capillary force of the CV-wick is 1670 N/m^3 at X = 10.5 mm;(d) the maximum capillary force of the GV-wick is 3720 N/m^3 at X = 10.5 mm; (e) Three-dimensional visualization of capillary force distribution for the CV-wick.

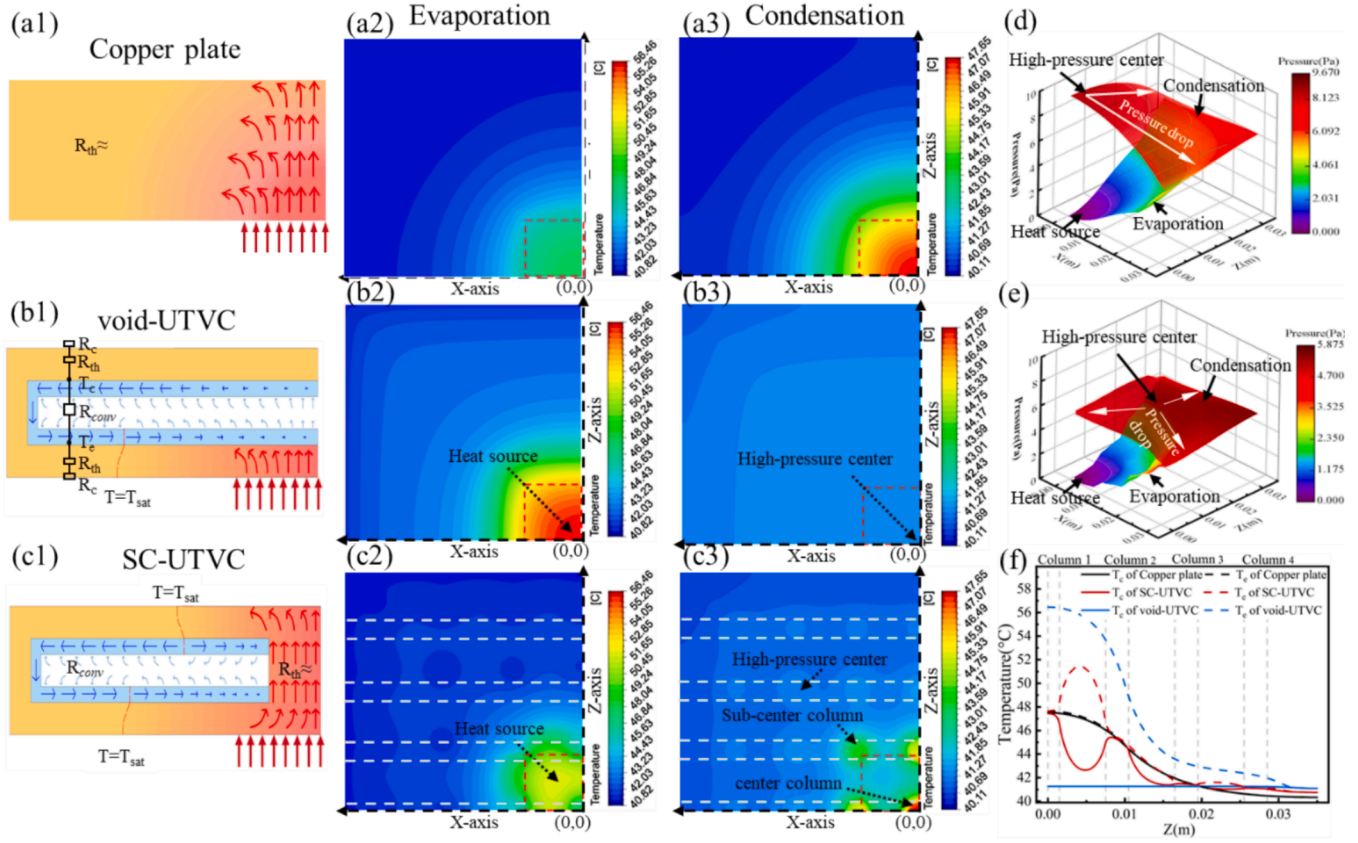


Fig. 3. (a1-c1) Schematic of the heat transfer mechanisms for copper plate, the void-UTVC, and SC-UTVC. (a2-c2) and (a3-c3) Their temperature distributions at the evaporation and condensation wick, respectively. (d,e) the gauge pressure distribution at the condensation and evaporation of the void-UTVC and SC-UTVC. (f) Their numerical temperatures for the condenser surface (y = 1 mm)and the evaporator surface (y = 0 mm), and the gray lines are the location of the corresponding support column.

Table 2
Note of name of UTVC.

Name of UTVC	Note
void-UTVC	UTVC without supporting columns
SC-UTVC	UTVC with supporting columns
D_{wc} = 1.5 mm	UTVC with 1.5 mm diameter, 9 mm spacing supporting columns
D_{wc} = 3 mm	UTVC with 3 mm diameter, 9 mm spacing supporting columns
D_{wc} = 4.5 mm	UTVC with 4.5 mm diameter, 9 mm spacing supporting columns
I_{wc} = 9 mm	UTVC with 1.5 mm diameter, 9 mm spacing supporting columns
I_{wc} = 6.75 mm	UTVC with 1.5 mm diameter, 6.75 mm spacing supporting columns
I_{wc} = 9 mm	UTVC with 1.5 mm diameter, 4.5 mm spacing supporting columns
W-UTVC	D_{wc} = 3 mm UTVC without grooves wick
CV-UTVC	D_{wc} = 3 mm UTVC with conventional grooves wick
GV-UTVC	D_{wc} = 3 mm UTVC with graded grooves wick

Fig. 3(a1-c1) illustrates three distinct heat transfer behaviors: single heat conduction through the copper plate, single-phase gas–liquid heat transfer in void-UTVC, and composite heat transfer in SC-UTVC. The thermal resistance network presented in Fig. 3(a1-c1) shows that the central support column actively contributes to heat transfer by providing a pathway with lower thermal resistance. The thermal conductivity of the central support column leads to additional heat accumulation on the condensing surface, which subsequently shifts the saturation temperature region of the vapor away from directly above the heat source. It is consistent with the temperature characteristics predicted by the Lee model [25]. Fig. 3(d, e) compares the gauge pressure distributions at the condensation and evaporation interfaces of void-UTVC and SC-UTVC. Introduction of the 3 mm diameter center column significantly reduces the pressure drop from 9.67 Pa to 5.875 Pa. Low-pressure areas in both SC-UTVC and void-UTVC are concentrated near the heat source, while high-pressure areas in void-UTVC occur at the vapor condensation center, and in SC-UTVC near the condensation center in the numerical model.

As shown in Fig. 3(e), the introduction of columns decreases both vapor and liquid pressure drops due to reduced saturation temperature and vapor mass fed by evaporation. Fig. 3(f) compares three types of numerical temperatures for the condenser surface (y = 1 mm) and the evaporator surface (y = 0 mm), with gray lines marking the locations of corresponding support columns. It is evident that SC-UTVC effectively reduces the temperature difference between evaporation and condensation compared to void-UTVC, with the highest temperature in SC-UTVC situated between the central support column and sub-central support columns. The copper flat plate exhibits smaller temperature differences in the vertical direction, but it struggles to transfer heat vertically away from the heat source, achieving only 5.3 % efficiency compared to the heat source center. These results underscore the enhanced heat transfer capabilities of center columns with high thermal conductivity in UTVCs, while highlighting minimal heat transfer through support columns in non-heat source areas.

To clarify the flow characteristics of increasing the column diameter in UTVCs, Fig. 4 depicts the comparison of liquid and vapor velocities between the void-UTVC and the SC–UTVC, respectively. In Fig. 4(a1), the liquid velocity at the condensation of void-UTVC is shown. The velocity increases along its flow direction due to liquid accumulation effects. Consequently, the region with lower liquid velocity is centered on the condensing surface, while higher velocities are observed at the sidewall edges. Fig. 4(a2) illustrates the evaporation of working liquid into vapor at the wick/liquid interface near the heat source. Vapor spreads from the core center outward, transferring heat from the source to the condensation surface. In the void-UTVC, vapor velocity is lowest

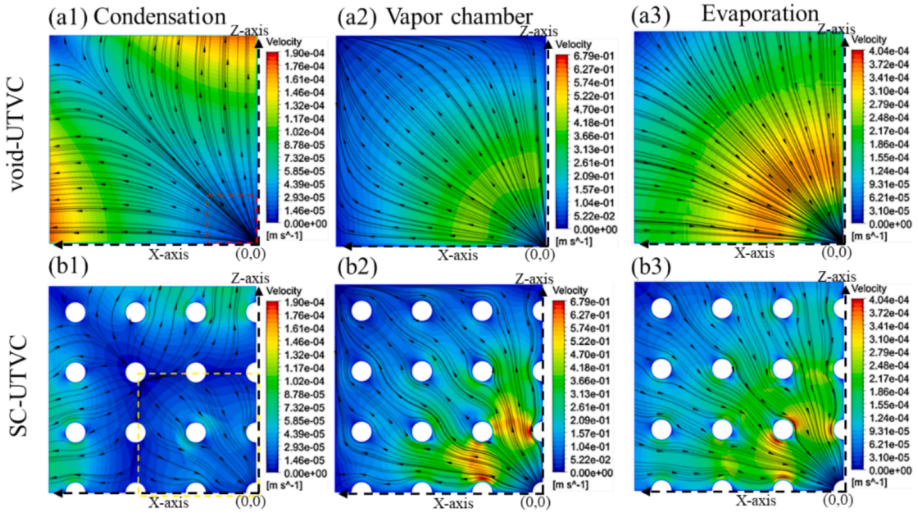


Fig. 4. Comparison of liquid/vapor velocity distribution between the void-UTVC and the SC–UTVC; (a1) and (b1) Liquid velocity on condensation surface; (a2) and (b2) Vapor velocity on profile A–A in vapor core; (a3) and (b3) Liquid velocity on evaporation surface.

at the vapor core center and side-wick, peaking at 0.68 m/s near the vapor core center. Fig. 4(a3) shows the distribution of liquid flow velocities at the evaporation interface, which mirrors the vapor velocity within the vapor core but in an inverse direction. Vapor velocities are generally higher than liquid velocities due to the significant density difference between the phases. At excessive temperatures at the heat source center, vapor velocity approaches zero, with condensation possible anywhere the temperature drops below saturation. Fluid tends to flow along paths of least resistance and shortest distance, highlighting the influence of steam saturation temperature distribution on reflux paths.

In Fig. 4(b1), the liquid velocity of the SC-UTVC exhibits a distinct distribution compared to the void-UTVC. Central columns in the heat source area enhance heat transfer and provide additional pathways for

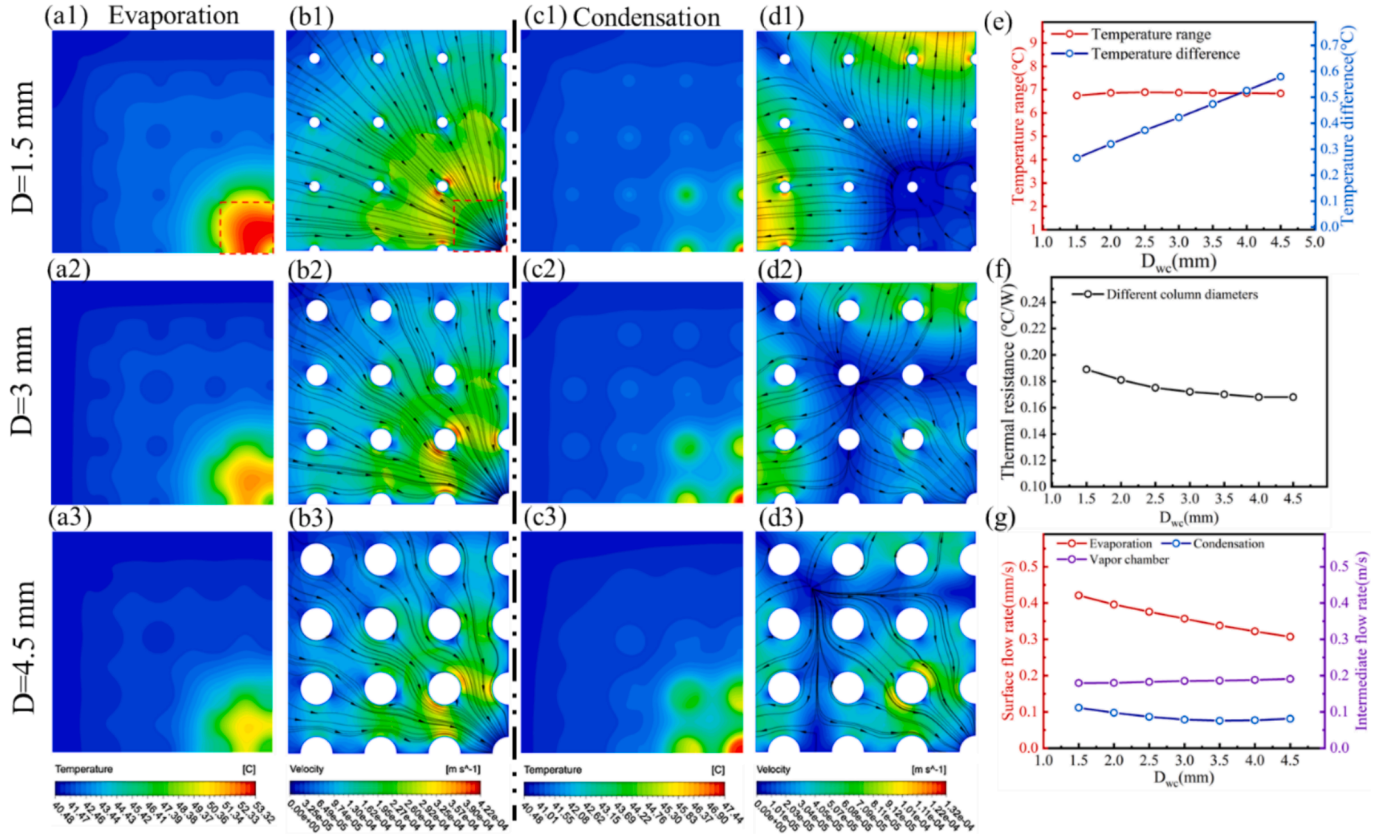


Fig. 5. Comparison of temperature and flow velocity distributions between SC–UTVC with different diameter support columns; (a1- a3) Temperature distributions of SC-UTVC with D_{wc} = 1.5 mm, 3 mm, and 4.5 mm at evaporation surface; (b1- b3) Liquid velocity distributions of three SC-UTVCs at evaporation surface;(c1- c3, d1-d3) Temperature and liquid velocity distributions of three SC-UTVCs at condensation surface; (e) Maximum temperature difference and temperature uniformity of SC-UTVC with different diameter support columns; (f) Thermal resistance of UTVC with different diameter support columns; (g)The average liquid/vapor velocity of SC-UTVC with different diameter support columns.

fluid circulation, which is different from auxiliary liquid circulation paths provided by the sintered cen-column reported by Lu [16]. This causes the liquid flow on the condensation surface to split into two regions based on minimal hydraulic resistance: one influenced by sidewalls and another by central columns, delineated by a clear reflux boundary (white dashed line). Fluid pumped by sidewall regions faces vapor resistance, resulting in higher velocities outside the reflux boundary. Fig. 4(b2) shows vapor flow influenced by support columns, which increases flow resistance, velocity, and alters flow direction within the vapor core. The highest velocity of 0.68 m/s is near the support column at the heat source edge, where fluid tends to evaporate initially. Inertia causes liquid to flow forward initially but decelerates rapidly, creating areas of minimal liquid and vapor velocity at the heat source center due to intense evaporation. As shown in Fig. 4(b3), liquid pumped by central columns alters evaporation locations, mass flow rates, and fluid directions on the evaporation surface, forming a multi-point evaporation area around the central column. Support columns thus serve multiple roles in vapor chambers, facilitating energy transfer via conduction and phase change processes, providing additional liquid flow paths to the evaporator, and structurally supporting the device. Therefore, the diameter and spacing of support columns are critical parameters in UTVC design.

3.3. Comparison of UTVC for different support column diameter

Fig. 5 (a-d) depicts the temperature and flow distributions at the evaporation and condensation surfaces of the SC-UTVC with $D_{wc} = 1.5$ mm, 3 mm, and 4.5 mm, respectively, all with a center-to-center distance of 9 mm. In Fig. 5(a1-a3), it is evident that increasing D_{wc} results in a decrease in the maximum temperature at the evaporation surface. This reduction is attributed to enhanced heat transfer facilitated by larger center columns in the heat source region. However, larger D_{wc} also increases fluid flow resistance and reduces the area available for phase change at the evaporation surface, as illustrated in Fig. 5(b1-b3). As shown in the Fig. 4(g), The flow velocities at evaporation surface for the SC-UTVC with $D_{wc} = 1.5$ mm, 3 mm, and 4.5 mm are 4.22 mm/s, 3.58 mm/s, and 2.98 mm/s, respectively. These velocities decrease as D_{wc} increases, reflecting the increased flow resistance.

At the condensation surface (Fig. 5(c1-c3)), the additional heat transferred by the center columns raises temperatures around the support columns, shifting the vapor saturation temperature from the heat source center towards the sidewalls. Consequently, increasing D_{wc} expands the area of higher temperature regions without significantly altering the maximum temperature. Fig. 5(d1-d3) depicts the mass flow rate distribution at the condensation surface. As the high-temperature region expands, the area exceeding the saturation temperature enlarges, gradually shifting the reflux boundary outward from the heat source center. Inside this boundary, where temperatures are higher, liquid mainly evaporates, resulting in reduced fluid velocity and flow rate. Outside the reflux boundary, lower temperatures promote condensation of the refluxing vapor, leading to increased vapor fluid velocity and flow rate. Consequently, increasing D_{wc} decreases the average flow rate at the condensation surface, while slightly increasing the vapor flow rate (Fig. 5(g)).

Fig. 5(e) illustrates that increasing the diameter of the support column does not elevate the maximum temperature range at the condensation surface. The temperature distribution of the SC-UTVC exhibits a progressively increasing non-uniformity with the increase in the diameter of the supporting column. Importantly, the average temperature difference across the UTVC decreases with larger support column diameters. In Fig. 5(f), the thermal resistances of SC-UTVC with $D_{wc} = 1.5$ mm, 3 mm, and 4.5 mm are measured as 0.187 ℃/W, 0.172 ℃/W, 0.168 ℃/W, respectively. Notably, beyond a support column diameter of 4 mm, thermal resistance shows no further decrease. These findings indicate that while increasing the diameter of the copper center column reduces the thermal resistance of the UTVC, it also increases the flow resistance of the working fluid. Therefore, optimizing the cross-sectional area through which the fluid passes become crucial under fixed conditions of additional heat flux.

3.4. Comparison of UTVC for different support column diameter interval

To minimize the disruption caused by substantial support columns on the flow of working fluid, a numerical analysis is conducted on the SC-UTVC with $D_{wc} = 1.5$ mm at different intervals. Fig. 6(a-d) illustrates temperature and flow distributions at the evaporation and condensation surfaces of the UTVC for support column intervals of $I_{wc} = 9$ mm, 6.75 mm, and 4.5 mm, respectively. Fig. 6(a1-a3) shows that reducing the support column interval will lower the maximum temperature at the evaporation surface, measuring 53.32 °C, 51.35 °C, and 50.37 °C, respectively. Adjusting the diameter or spacing of support columns increases the total area of support columns within the heat source region, thereby reducing the maximum temperature at the evaporation surface. Conversely, support columns positioned farther from the heat source region have minimal impact on this temperature reduction. The flow resistance increases with decreasing spacing between support columns, as depicted in Fig. 6(b1-b3), with average flow velocities at the evaporation surface measured at 0.42 mm/s, 0.35 mm/s, and 0.25 mm/s, respectively (Fig. 6(g)). Smaller spacing introduces more additional heat transfer points, resulting in poorer temperature uniformity at the condensation surface, as observed in Fig. 6(c1-c3). The distribution of vapor saturation temperature is influenced by the cumulative heat transfer from the center columns, causing the reflux boundary line to gradually shift towards the sidewalls as spacing decreases (Fig. 6(d1-d3)).

Compared to increasing support column diameter, reducing support column spacing can simultaneously enhance the hydraulic performance of the UTVC and provide an additional thermal pathway between evaporation and condensation. Fig. 6(e) illustrates that the maximum temperature difference in the UTVC increases with wider support column spacing, while temperature uniformity decreases. The thermal resistance of the SC-UTVC with $I_{wc} = 9$ mm, 6.75 mm, and 4.5 mm are 0.189 ℃/W, 0.168 ℃/W, 0.158 ℃/W, respectively in Fig. 6(f).

3.5. Comparison of UTVC for different V-type microgroove

Fig. 7 illustrates the temperature and liquid velocity distributions of W-UTVC (the SC-UTVC without wicks), CV-UTVC, and GV-UTVC at the evaporation and condensation surfaces. As depicted in Fig. 7(a1-a3), the maximum temperatures at the evaporation surface for CV-UTVC and GV-UTVC are slightly lower than those of W-UTVC, with minimal change in their temperature distribution. The average temperatures of W-UTVC, CV-UTVC, and GV-UTVC are 41.64°C, 41.59°C, and 41.49°C, respectively (Fig. 7(e)). The impact of capillary forces generated by the axial grooves on the temperature distribution at the evaporation surface appears to be non-directional. Fig. 7 (b1-b3) shows the liquid velocity distribution at the evaporation surface affected by the capillary force, where fluid velocity is directed towards the axial grooves.

Examining the flow rate distributions at the evaporation surface with coordinates X = 3 mm, Y = 0.17 mm (Fig. 7(g)), the maximum flow rate of W-UTVC is 1.63 mm/s within the heat source region, whereas those of GV-UTVC and CV-UTVC are increased by 12.2 % and 9.2 %, respectively. However, the flow rates of all three UTVC configurations are comparable in areas distant from the heat source. At the condensation surface, the temperature distribution is shown in Fig. 7 (c1-c3), where the average temperatures of W-UTVC, CV-UTVC, and GV-UTVC are 40.99°C, 40.98°C, and 40.93°C, respectively (Fig. 7(f)). Computing the thermal resistance of the three UTVC configurations at both condensation and evaporation reveals that GV-UTVC experiences an 8.13 % reduction and CV-UTVC a 5.8 % reduction compared to W-UTVC.

Although condensation is not influenced by the capillary force model, the gas–liquid circulation is notably accelerated by the increased

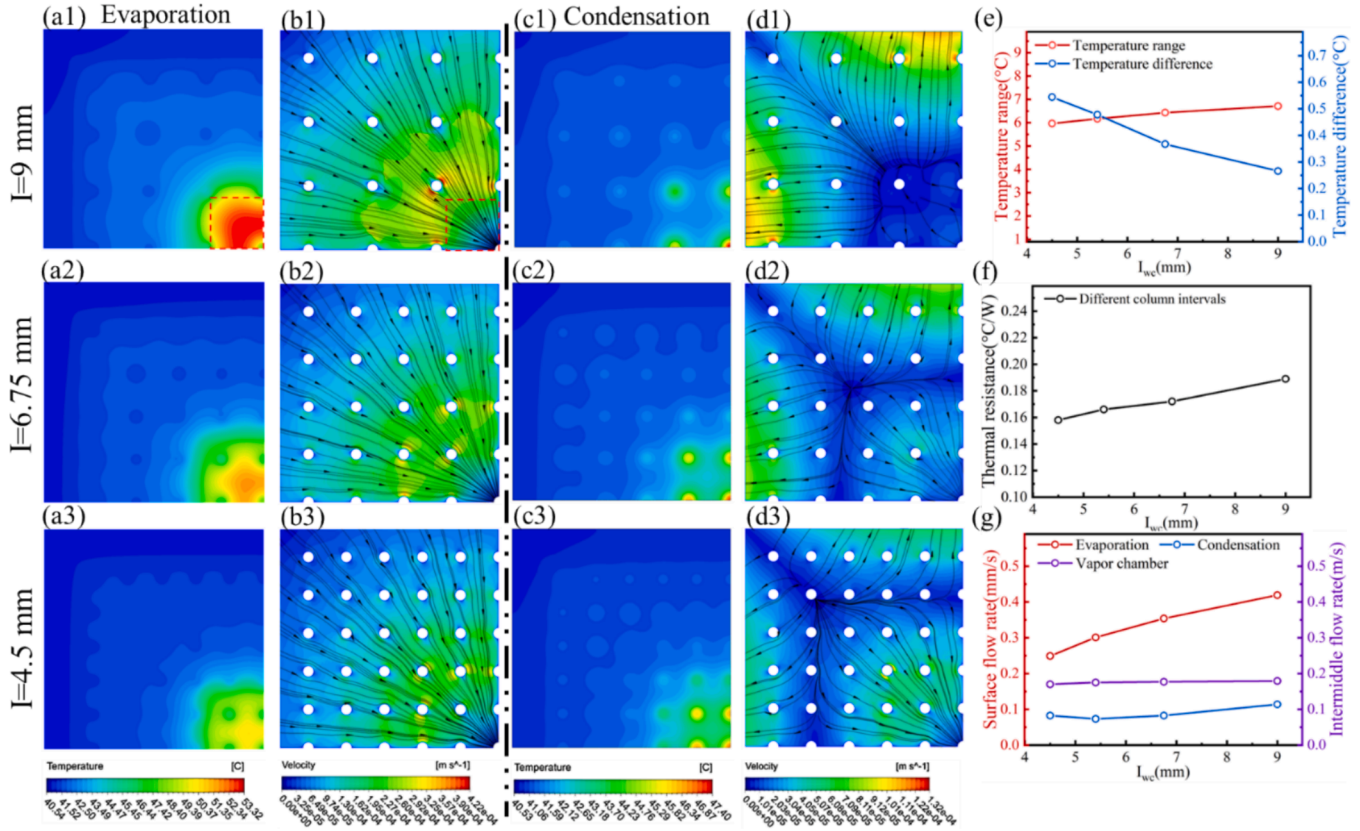


Fig. 6. Comparison of temperature and flow velocity distributions between the SC–UTVC with different interval support columns; (a1- a3) Temperature distributions of the SC-UTVC with I_{wc} = 9 mm, 6.75 mm, and 4.5 mm at evaporation surface; (b1- b3) liquid velocity distributions of three SC-UTVCs at evaporation surface;(c1- c3) and (d1- d3) temperature and liquid velocity distributions of three SC- UTVCs at condensation surface; (e) maximum temperature difference and temperature uniformity of UTVC with different interval support columns; (f) thermal resistance of SC-UTVC with different interval support columns; (g) the average liquid/vapor velocity of SC-UTVC with different interval support columns.

flow velocity at the evaporation surface due to capillary forces. Consequently, the maximum flow velocity of CV-UTVC slightly increases compared to W-UTVC, especially with the introduction of the GV-wick, further enhancing reflux velocity (Fig. 7 (d1-d3)). The simulation results demonstrate that optimizing the support column structure and implementing gradient-distributed groove capillary forces in the heat source region can significantly enhance the heat transfer performance of UTVCs.

4. Experiment validation

To verify the feasibility of the mathematical model, the graded microgroove wick UTVC is fabricated using ultrafast laser technology, and an independent heat transfer test device is constructed for performance evaluation in this study. In order to analyze the magnitude of the model error and to verify the reasonableness of the model, CV-UTVC is used in this study as a comparison with the simulation results of the UTVC with the same structure. The phase transition surface temperature curves as a function of heating power, obtained from both experimental and simulation results, are shown in Fig. 8(a). The error between the simulated and experimental results for the phase change surface temperature increases with heating power because the actual UTVC liquid film thickness decreases as evaporation increases, whereas the simulation assumes a constant liquid film thickness. At a heating power of 45 W, the errors in the maximum temperatures at the evaporation and condensation surfaces are 9.35 K and 1.86 K, respectively, with relative errors of 16.4 % and 4.16 %, respectively. The comparison of errors with reported papers are shown in Table S3. Since the thermal resistance of the UTVC is primarily determined by the liquid film thickness, the experimental results show a decrease in thermal resistance with increasing heating power, while the simulation results show a constant thermal resistance, as depicted in Fig. 8(b).

The phase transition surface temperature and thermal resistance curves of the UTVC at different filling rates are shown in Fig. 8(c) and (d), respectively. In the simulation results, the liquid film thickness increases with the liquid filling rate, leading to an increase in the thermal resistance of the UTVC. In contrast, the experimental results show that the thermal resistance of the UTVC is mainly influenced by the internal vapor pressure drop and capillary force, resulting in an optimal liquid filling rate. In this study, the optimal liquid filling rate is 15 %. Although the temperature errors associated with the proposed numerical scheme are not exceptional when compared with pioneering works, these temperature errors are still reasonable and acceptable in a typical engineering context.

To verify the feasibility of the mathematical model, the graded microgroove wick UTVC is fabricated using ultrafast laser technology, and an independent heat transfer test device is constructed for performance evaluation in this study. The 3D contour map and SEM image of the V-type groove are depicted in Fig. 9(a) and (b). Fig. 9(c) presents the thermal resistance profiles against input power for $D_{1.5}$- UTVC and D_3- UTVC. The distribution of supporting columns in $D_{1.5}$- UTVC is shown in Fig.S2, and the structural parameters for all UTVCs used in the experiment are detailed in Table 3. Experimental findings indicate that the thermal resistance of $D_{1.5}$- UTVC is reduced by up to 7.6 % compared to that of D_3- UTVC when the heating power is below 20 W. However, at heating powers exceeding 20 W, rapid drying out of the liquid mass inside the $D_{1.5}$- UTVC occurs, leading to a sharp increase in thermal resistance. While the larger diameter center support column can lower

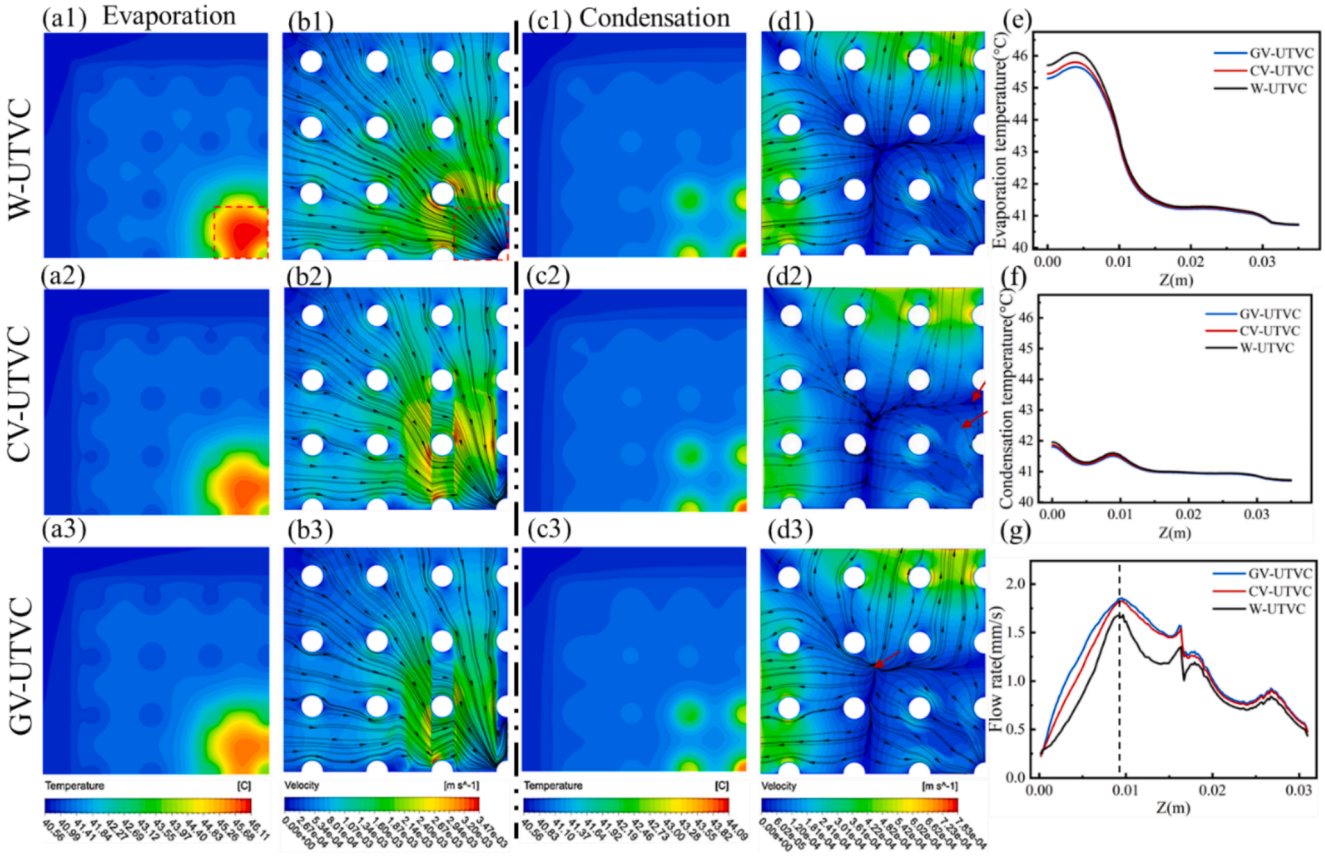


Fig. 7. Comparison of temperature / liquid velocity distribution of W-UTVC, V-UTVC and GV-UTVC; (a1- a3) Temperature distributions of UTVCs at evaporation; (b1- b3) Liquid velocity distributions of UTVCs at evaporation; (c1- c3)Temperature distributions of UTVCs at condensation; (d1- d3)Liquid velocity distributions of UTVCs at condensation; (e)Temperature profile of UTVCs at evaporation from the center to the edge; (f) Temperature profile of UTVCs at condensation from the center to the edge; (g) Flowrate of UTVCs from the center to the edge.

the maximum temperature at the evaporating surface, it also diminishes the phase change area within the heat source region, increasing susceptibility to drying out and significantly reducing the UTVC's heat load capacity.

Fig. 9(d) illustrates the thermal resistance curves of the prepared GV-UTVC and CV-UTVC as a function of heating power at a liquid filling rate of 40 %. Below 25 W of heating power, GV-UTVC exhibits a 29.3 % lower thermal resistance compared to CV-UTVC. However, beyond 25 W, GV-UTVC shows a substantial increase in thermal resistance due to rapid drying out of the liquid in the heat source area. The experimental results demonstrate that optimizing the center columns in the heat source region and implementing a gradient capillary force microgroove can effectively reduce the thermal resistance of the UTVC, consistent with numerical simulations. Nonetheless, the model used in this study assumes a fixed gas–liquid interface, neglecting variations in liquid film thickness. Therefore, it does not predict the significant reduction in drying out resistance observed in the optimized UTVC, indicating some limitations in the simulation results.

5. Conclusions

In this study, a three-dimensional steady state fluid heat transfer model is developed. The model evaluates flow resistance and enhanced heat transfer from the central column, analyzing the influence of supporting column diameter, supporting column interval, and type of microgroove on temperature and flow velocity distributions. Following these numerical analyses, ultra-thin vapor chambers with gradient microgroove wicks were fabricated using ultrafast laser technology, and the discrepancies between experimental and numerical results under various conditions were assessed. The main findings are as follows:

(1) The central column enhanced heat transfer and disturbed the temperature uniformity, leading to a shift of the saturation temperature towards the wall, creating an additional path for liquid circulation.
(2) The graded microgroove enhances the reflux velocity and overall heat transfer performance of ultra-thin vapor chamber.
(3) Experimentally, the thermal resistance of ultra-thin vapor chamber with graded microgrooves is 29.3 % lower than that with conventional microgrooves, with a maximum relative error of 4.16 % between simulated and experimental condensation temperatures.

Conflict of Interest Statement

In submitting this manuscript to your esteemed journal, we, the undersigned authors, hereby declare that there are no financial or personal relationships that could inappropriately influence (bias) our work as described in the manuscript entitled "[**Thermal performance analysis of supporting column ultra-thin vapor chamber with graded microgroove]** ".

Specifically, we declare that:

1. No author has a financial interest in any company or other entity that has a direct financial interest in the subject matter of the manuscript.
2. No author has received any payment or other benefits from a third party for any aspect of the work related to the manuscript.

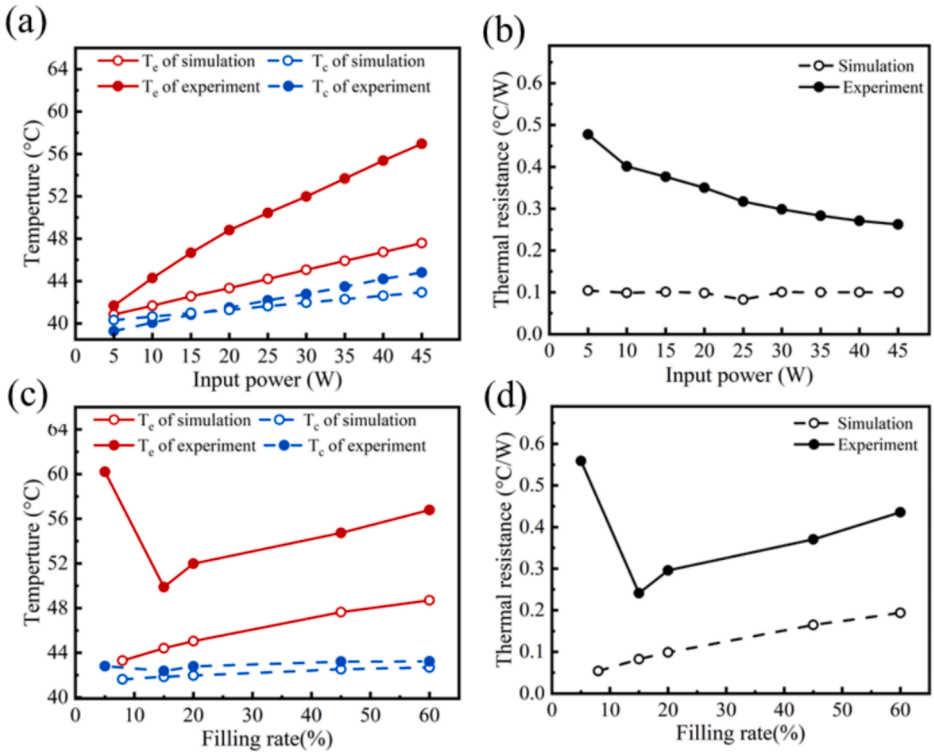


Fig. 8. (a)The curves of evaporation and condensation temperatures of UTVC with heating power for experimental and simulated results; (b)Thermal resistance of UTVC with heating power for experimental and simulated results. (c)The curves of evaporation and condensation temperatures of UTVC at different liquid filling rates for experimental and simulated results; (d) Thermal resistance of UTVC at different liquid filling rates for experimental and simulated results.

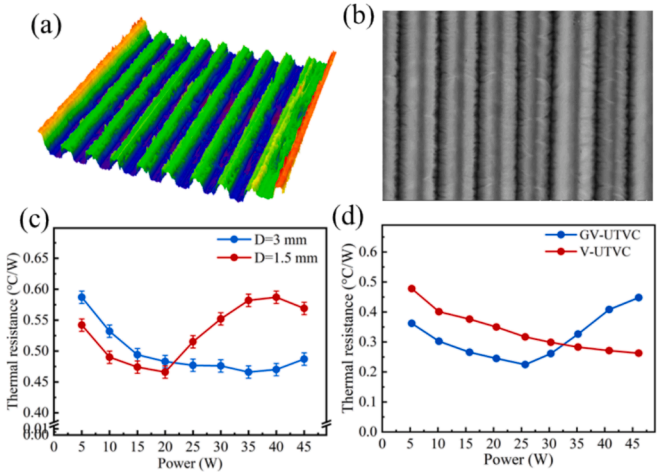


Fig. 9. (a)The 3D confocal image of V-type groove; (b) SEM image of V-type groove; (c) Comparison of experimental and simulation results with different diameter at heat load; (d) Comparison of experimental and simulation results with different V-type groove at heat load.

Table 3
Structure parameters of all UTVCs in experiment.

Name	Diameter (D_{wc})	Interval (I_{wc})	Type of groove wick	filling rate
D_3- UTVC	3 mm	9 mm	Conventional V-type	15 %
$D_{1.5}$- UTVC	1.5 mm	6.75 mm	Conventional V-type	15 %
GV-UTVC	3 mm	9 mm	Gradient V-type	15 %
CV-UTVC	3 mm	9 mm	Conventional V-type	15 %

3. No author holds any stocks or shares in a company that may gain or lose financially from the publication of the results contained in the manuscript, either now or in the future.
4. No author has engaged in any consultancy or speaking activities that may be seen as a conflict of interest in relation to the work described in the manuscript.
5. No author has any other relationships or activities that could appear to have influenced the submitted work.

We confirm that the work described in the manuscript has been conducted in accordance with the highest standards of ethical conduct and that all data have been subjected to rigorous and appropriate review. We pledge to promptly notify the journal if any such conflicts arise in the future.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.applthermaleng.2024.124578.

Data availability

Data will be made available on request.

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